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Dynamic Stochastic General Equilibrium and Business Cycles

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Research Track Master

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Chapter 1

An Introduction to Dynare Programming

Content of the Handout

Handout Objectives

- Introducing to the simulation of a very simple business cycle model.
- Solving any theoretical model, find its steady state, log-linearise it and plot its IRFs.
- Computing second moment statistics by hand using artificial series.

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1. Theoretical foundations and resolution of a simple DSGE model

The term dynamic stochastic general equilibrium model defines a class of models widely used in macro which have the following features:

1. They describe and characterize the comovement and evolution of (random) economic variables over time;
2. They are normally based on microfoundations and they are based on the hypothesis that markets clear through adjustment of prices and quantities;
3. They rely on random fluctuations in technology, preferences and other exogenous sources as the primary impulse for the movement of the economic variables over time.

In this lecture, we present a simple Real Business Cycle (RBC) model. The lecture strongly relies on the course of Iacoviello (2009)
(corresponding webpage: https://www2.bc.edu/ iacoviel/teach/0809/EC751.html).

1.1 An extensive presentation of the model

Our economy is characterized by a representative household which consumes, produces goods, accumulates physical capital involved in the production. There is no labour in this economy as household's labour supply is assumed to be inelastic to macroeconomic fluctuations while technology does not involve hours worked. Households are immortal and infinite, the economy is populated by an infinite number of $j \in [0,1]$ distributed on an unitary contiuum.

The standard assumption in macroeconomics is that consumption delivers utility through a utility function, here we assume utility is logarithmic in consumption:

$$\mathcal{U}\left(C_{jt}\right) = \log\left(C_{jt}\right)$$

Under these preferences, the houhold gets more utility whenever consumption is higher ($\mathcal{U}'\left(C_{jt}\right) > 0$), but that consumption runs into diminishing returns ($\mathcal{U}''\left(C_{jt}\right) < 0$), *i.e.* diminishing marginal utility. That is, each additional unit of consumption raises utility by a smaller and smaller amount, this assumption is a key pattern of DSGE models to explain consumption fluctuations over time.

Since our problem is dynamic, the welfare index of the representative household reads as follows:

$$\mathcal{W}_t = \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^{\tau} \mathcal{U}\left(C_{t+\tau}\right)$$

The welfare index is assumed to be defined as the expected sum of utilities where $0 \leq \beta < 1$ denotes the discount factor. For $\beta = 1$, the household is perfectly patient and does not discount the future, *i.e.* consuming now or later provides exactly the same utility ($\beta^0 \mathcal{U}\left(\bar{C}\right) = \beta^1 \mathcal{U}\left(\bar{C}\right) = \beta^2 \mathcal{U}\left(\bar{C}\right)$). For any value of $\beta < 1$, the household is impatient, the lower β is, the more impatient the household is ($\beta^0 \mathcal{U}\left(\bar{C}\right) > \beta^1 \mathcal{U}\left(\bar{C}\right) > \beta^2 \mathcal{U}\left(\bar{C}\right)$). Micro evidence reveals households are impatient, so the discount factor should lie between 0 and 1.

Expression $\mathbb{E}_t$ denotes the expectation operator. In a rational expectation equilibrium, agents can form expectations about the future states of the economy at no cost. However in a stochastic environment, the expectation of $\mathbb{E}_t\left\{X_{t+1}\right\}$ is assessed with all information available in t to agents, meaning that any shock that will occurs between t and $t+1$ will not be anticipated: a gap between the expectation of $\mathbb{E}_t\{X_{t+1}\}$ and its realization X_{t+1} is observed, thus creating a source of fluctuation.

The households faces a (intertemporal) budget constraint:

$$C_{jt} + I_{jt} = Y_{jt} \tag{1}$$

The total income Y_{jt} is spent on consumption goods C_{jt} and investment goods I_{jt}. An intertemporal budget constraint shows what consumption options are available between the present and future. The constraint can be used to determine the future value of present income or the present value of future income. This constraint binds each period, *i.e.* households expenditures equal her income for any value of t. Here, the budget constraint can also be interpreted as the definition of the GDP using expenditure approach (neglecting both stocks and the trade balance).

The law of motion of capital reads as in the Solow (1956)'s model:

$$I_{jt} = K_{jt} - (1-\delta)\, K_{jt-1} \tag{2}$$

Investment is determined by the gap between current capital K_{jt} and depreciated capital $(1-\delta)\, K_{jt-1}$. In the latter expression, the parameter $0 \leq \delta \leq 1$ denotes the depreciation rate of capital.

The standard Cobb-Douglas technology is determined by:

$$Y_{jt} = A_t K_{jt-1}^{\alpha} \tag{3}$$

where $0 < \alpha \leq 1$ is the share of capital involved in the production process of new goods Y_{jt} while A_t is a productivity shock which temporary affects the society's ability to produce goods, it affects all the households in our economy in the same way. The production combines technology and capital K_{jt-1} bought in $t-1$ as capital requires one period to be settled (the so called time-to-build in the RBC literature). Concerning the productivity shock, it is assumed to follow a simple AR(1) process given by:

$$\log(A_t) = \rho \log(A_{t-1}) + \eta_t, \tag{4}$$

where $0 \leq \rho < 1$ is the autocorrelation coefficient of the shock and η_t is an innovation distributed normally with zero mean and finite variance σ^2 such that $\eta_t \sim \mathcal{N}(0, \sigma^2)$.

1.2 Solving the optimization problem

Substituting Equation 2 in Equation 3, the problem boils down into:

$$\max_{C_{jt}, K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^{\tau} \log(C_{jt+\tau})$$
$$s.t. : C_{jt} + K_{jt} = A_t K_{jt-1}^{\alpha} + (1-\delta)\, K_{jt-1}$$

To get the optimality conditions, we solve the optimisation problem using a (dynamic) lagrangian:

$$\mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^{\tau} \left[\log(C_{jt+\tau}) - \lambda_{t+\tau}\left[C_{jt+\tau} + K_{jt+\tau} - A_{t+\tau} K_{jt-1+\tau}^{\alpha} - (1-\delta)\, K_{jt-1+\tau}\right]\right] \tag{5}$$

where λ_t denotes the lagrange multiplier on the budget constraint. The budget constraint is included in the discounted sum as this constraint binds each period.

First order conditions which solve problem 5 read as follows:

$$(C_{jt}) : \frac{1}{C_{jt}} - \lambda_t = 0$$
$$(K_{jt}) : -\lambda_t + \beta \mathbb{E}_t \lambda_{t+1}\left[\alpha A_{t+1} K_{jt}^{\alpha-1} + (1-\delta)\right] = 0$$

Optimal intertemporal allocation occurs when the intertemporal indifference curve and the intertemporal budget constraint are tangent. This condition can be found by replacing the lagrange multiplier to obtain the Euler equation:

$$\frac{1}{C_{jt}} = \mathbb{E}_t \frac{1}{C_{jt+1}} \beta \left[\alpha A_{t+1} K_{jt}^{\alpha-1} + (1-\delta) \right] \tag{6}$$

The Euler equation describes the intertemporal optimal path of consumption over the business cycles. In this set-up, current consumption C_{jt} is negatively driven by the marginal product of capital. The household faces a trade-off between consuming now, or reporting its consumption later through capital buying. If the marginal product of capital increases (*i.e.* the return from holding one unit of capital between t and t+1 rises), the household prefers buying physical capital now and reports his consumption later. Equation 6 can also be interpreted as the no-arbitrage condition between consumption and capital which is satistified when the marginal utility of consumption equals the marginal product of physical capital. Regarding the interaction between consumption in C_{jt} and C_{jt+1} as written in the Euler Equation 6, it will be explained in Chapter 3 through the Friedman's permanent income theory with a full fledge Real Business Cycle model.

Finally, we aggregate all the households populating our economy simply through:

$$C_t = \int_0^1 C_{jt} \mathrm{d}j \quad \text{and } K_t = \int_0^1 K_{jt} \mathrm{d}j$$

Our economy is characterized by a set of 3 equations:

$$\frac{1}{C_t} = \mathbb{E}_t \left\{ \frac{1}{C_{t+1}} \beta \left[\alpha A_{t+1} K_t^{\alpha-1} + (1-\delta) \right] \right\} \tag{7}$$

$$C_t + K_t = A_t K_{t-1}^{\alpha} + (1-\delta) K_{t-1} \tag{8}$$

$$\log(A_t) = \rho \log(A_{t-1}) + \eta_t \tag{9}$$

Finally to obtain a stationary system, we set a transversality condition which avoids ponzi games such that households could accumulate indefinitively capital. This condition states that the present discounted value of capital has to be equal to zero; that is $\lim_{\tau \to +\infty} \beta^\tau C_{t+\tau}^{-1} K_{t+\tau} = 0$.

1.3 Theoretical insights behind the Euler Equation

The Euler equation provides the basis of the neoclassical consumption model. As shown previously, individuals choose consumption at each point in time to maximize a lifetime utility function that depends on current and future consumption. Households recognize that income in the future may differ from income today, and such differences influence consumption today.

The essence of the neoclassical model lies in the hability of households to postpone consumption in the future through a financial asset. This asset can either be risk-free, as for government bonds and money, or risky, as for shares or physical capital. Here, the households can postpone consumption through the purchase of units of capital. However, this asset is not a free lunch as it depreciates each period at a fixed coefficient δ, and its expected returns are risky as they depend on future realizations of shocks (involving uncertainty). To underline these two aspects of this financial asset, let us consider the Euler equation:

$$\mathcal{U}'(C_t) = \beta \mathbb{E}_t \left\{ \mathcal{U}'(C_{t+1}) \left[\alpha A_{t+1} K_t^{\alpha-1} + (1-\delta) \right] \right\},$$

where $\mathcal{U}'(C_t)$ is the marginal utility of consumption. Letting r_t denote the gross return of physical capital:

$$r_t = \alpha A_t K_{t-1}^{\alpha-1} + (1-\delta) = \alpha \frac{Y_t}{K_{t-1}} + (1-\delta) = mpk_t + (1-\delta),$$

where mpk_t is the marginal product of physical capital. So basically, when the household holds one unit of capital, she gets in terms of return the marginal product of physical capital less the cost of depreciation.

The Euler thus reads now as:

$$\mathcal{U}'(C_t) = \beta \mathbb{E}_t \{\mathcal{U}'(C_{t+1}) r_{t+1}\} \tag{10}$$

The left hand side of this equation reports the cost of buying one unit of physical capital today (priced in terms of present marginal utility), and the right hand side is the expected return of holding this unit between t and $t+1$ (priced in terms of future marginal utility). This pricing in terms of utility allows to evaluate in terms of consumption equivalents each unit of capital purchased by the household. Because of risk aversion in the utility function, the household purchases physical capital untill the future return of physical capital reaches its present marginal cost. As expected in this standard model, the equilibrium is characterised by the standard marginality conditions—the marginal product of capital equals r_{t+1} and the intertemporal marginal rate of substitution of future for present consumption $\mathcal{U}'(C_t)/\mathcal{U}'(C_{t+1})$:

$$\frac{\mathcal{U}'(C_t)}{\mathbb{E}_t\{\mathcal{U}'(C_{t+1})\}} = \beta \mathbb{E}_t \{r_{t+1}\} \tag{11}$$

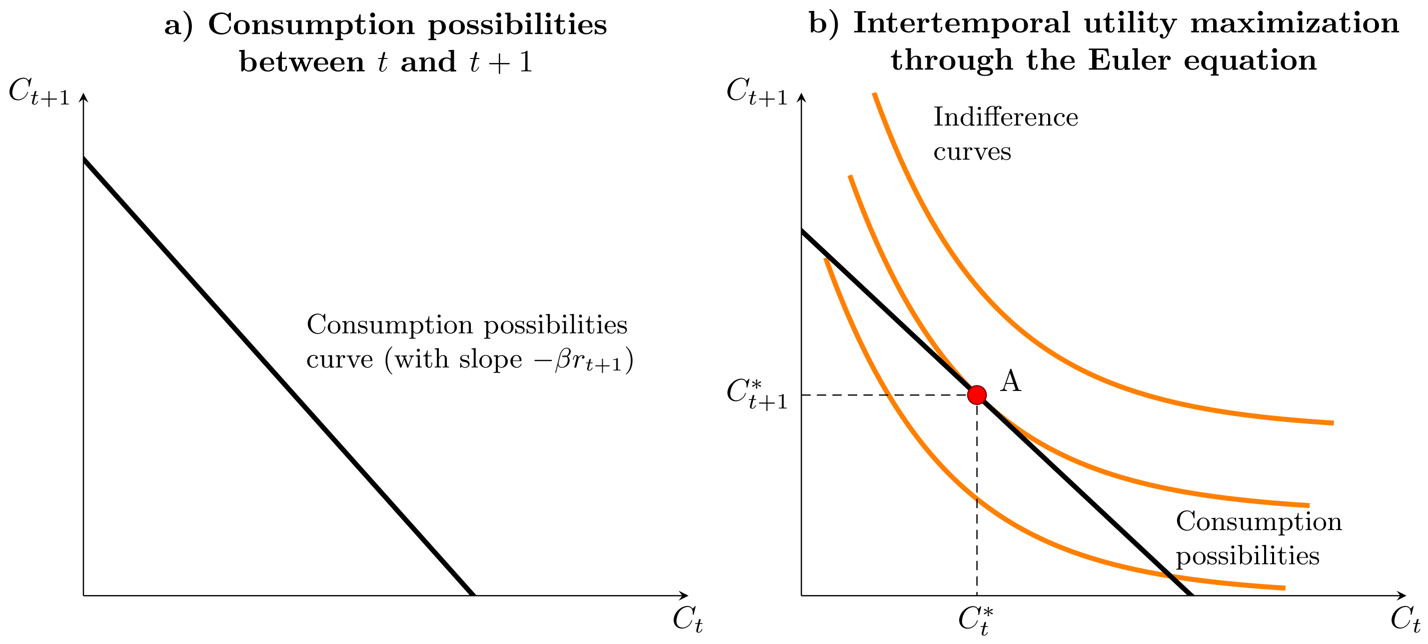

Figure 1: Optimization scheme for the representative household

Figure 1 reports the joint-determination of the optimal level of financial assets and consumption. In the left panel, we show the consumption possibilities (C_t, C_{t+1}) which are offered by financial markets in the economy (and implicitely here by the production capacities of the economy). The slope of this curve,$-\beta \mathbb{E}_t\{r_{t+1}\}$, is decreasing because future consumption is on the vertical axis. Given the real rate on physical capital, the household determines the level of present and future consumption (C_t, C_{t+1}) maximizing its utility given the consumption possibilities offered by financial markets. In equilibrium A, the household find the unique possible path of consumption maximizing her utility intertemporally and is called *the consumption point*.

Finally in autarky (*i.e.*, with no trading partners), the general equilibrium is jointly determined by the equilibrium on financial and goods markets, as the optimal quantity of physical capital determines both the level of savings and production, and in turn of consumption in the economy. In Figure 2, the left panel reports the optimal determination of output, given the production technology offered by the Cobb-Douglas

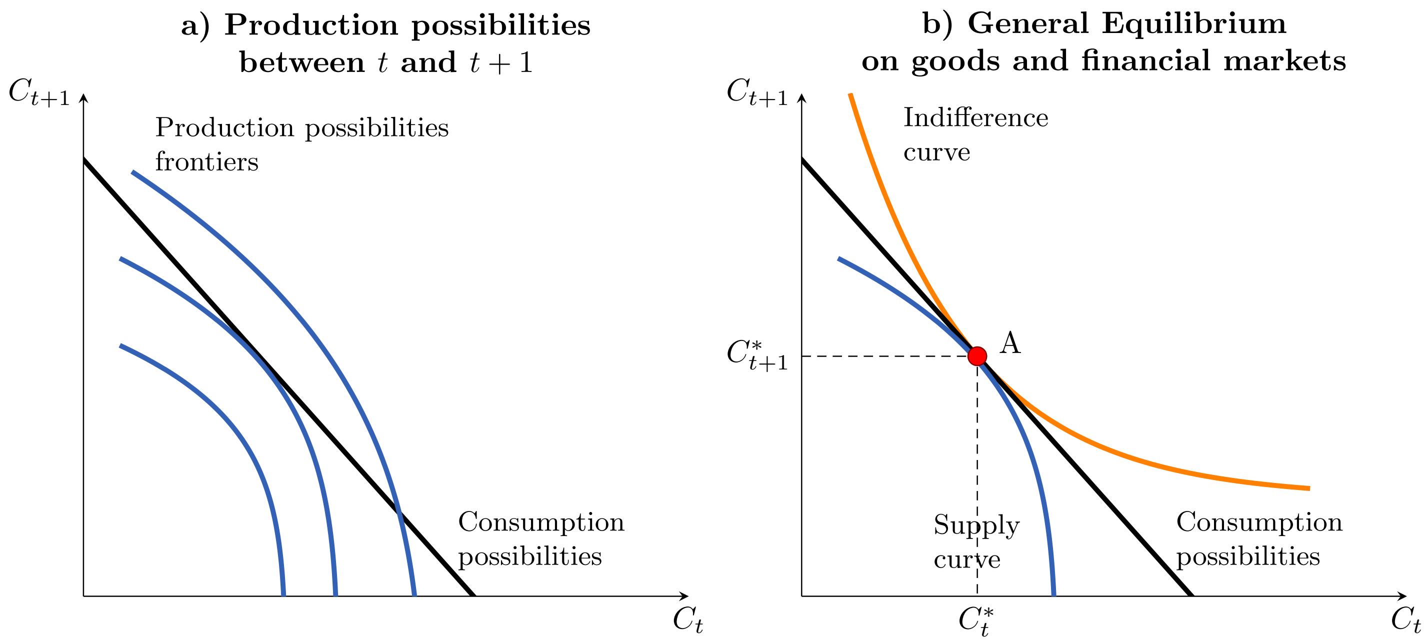

Figure 2: General equilibrium in a closed economy setup

functional form of the supply curve. Since supply is limited by the demand from consumers, the optimal allocation is given by the unique *production point* in equilibrium B in the right figure. In a closed economy setup, the consumption and the production points are the same, however in an open economy setup, the trade balance allows to have a different level of production and consumption.

1.4 Solving the model in a rational expectation equilibrium

Our 3 equation model is a non-linear system with rational expectations. In compact notation, the system can be represented as follows:

$$\mathbb{E}_t \left[f_\theta \left(y_{t+1}, y_t, y_{t-1}, \epsilon_t \right) \right] = 0 \tag{12}$$

where:

- y is the vector collecting all the endogenous variables;
- ϵ is the vector collecting the exogenous stochastic shocks;
- θ is the vector collecting the deep parameters of the model;
- $f_\theta(.)$ is the model;

The solution to a model like this is normally a set of functions relating controls to states such that our 3 equations are satisfied at all times. For most of the problems, we are interested in the response of y_t after the realization of the shocks ϵ_t. For instance, our solution for y_t is a function:

$$y_t = g\left(y_{t-1}, \epsilon_t\right)$$

such that Equation 12 holds. The problem is that there is an infinite number of values of y_{t-1} and ϵ_t and an infinite number of values of y_t to find. Hence, with a few rare exceptions what we do to solve DSGE models is to approximate $g(.)$ by a linear (or quadratic) function around a well defined point (here, the steady state).

The approximation is usually performed up to first order in terms of deviation from the steady state. For instance, let us consider in our model capital supply decisions K_t between t and $t+1$, with K_0 given, which is determined by the Euler first order condition denoted $v(.)$ such that $(K_t, K_{t+1}, K_{t+2}, ...) = 0$. The solution of the problem can be rewritten in a recursive fashion given by $K_{t+1} = g(K_t, \epsilon_t)$ which describes the intertemporal equilibium $\forall t$. Using the recursive expression, our first order condition on capital now reads as follows: $v(K_t, g(K_t), g(g(K_t)), ...) = 0$, which implies an infinite number of equations (one equation for each K_t) for an infinite number of variables (one solution $g(.)$ for each K_t).

The standard way to bound the problem is to approximate the function $v(.)$ by a log-linear function around the steady state using the method of Uhlig et al. (1995). Under particular conditions defined by Blanchard and Kahn (1980), our linear rational expectations model has a stable and convergent intertemporal equilibrium.

More precisely, we usually take the following steps to solve a model:

1. Find the steady state of the model;
2. (log-)Linearize around the steady state;
3. Write the Dynare file.

1.5 Steady-state

To get the steady state, we consider our set of equations assuming $\bar{X} = X_{t-1} = X_t = X_{t+1}$ where $\bar{X}$ denotes the steady state of the random variable X_t. We obtain:

$$\begin{aligned}
\frac{1}{\bar{C}} &= \frac{\beta}{\bar{C}}\left[\alpha\bar{A}\bar{K}^{\alpha-1} + (1-\delta)\right] \\
\bar{C} + \bar{K} &= \bar{A}\bar{K}^{\alpha} + (1-\delta)\bar{K} \\
\log\left(\bar{A}\right) &= \rho\log\left(\bar{A}\right)
\end{aligned}$$

while normalizing to one the technology $\bar{A} = 1$ clears the shock process as $\log\bar{A} = 0$ (recall that the innovation is zero mean such that $E(\eta_t) = 0$).

After some simple algebra, the steady state of $\bar{K}$ and $\bar{C}$ is:

$$\bar{K} = \left[\left(\frac{1}{\beta} - (1-\delta)\right)\frac{1}{\alpha}\right]^{\frac{1}{\alpha-1}} \tag{13}$$

$$\bar{C} = \bar{K}^{\alpha} - \delta\bar{K} \tag{14}$$

1.6 Log-linearization

All the tricks regarding the log-linearization are detailed in Appendix B.

Log-Linearization of the Euler equation:

$$\underbrace{\mathbb{E}_t\left\{C_{t+1}\right\}/C_t}_{x_{1,t}} = \underbrace{\beta\alpha\mathbb{E}_t\left\{A_{t+1}\right\}K_t^{\alpha-1}}_{x_{2,t}} + \underbrace{\beta(1-\delta)}_{x_{3,t}}$$

$$\begin{aligned}
&\text{with } x_{1,t} \to \frac{\bar{C}}{\bar{C}}\left(\mathbb{E}_t\left\{\hat{c}_{t+1}\right\} - \hat{c}_t\right) \\
&\text{and } x_{2,t} \to \beta\alpha\bar{A}\bar{K}^{\alpha-1}\left(\mathbb{E}_t\left\{\hat{a}_{t+1}\right\} + (\alpha-1)\hat{k}_t\right) \\
&\text{and } x_{3,t} \to 0 \quad \text{(as it is a constant term)}
\end{aligned}$$

Then, current consumption is determined by:

$$\hat{c}_t = \mathbb{E}_t\left\{\hat{c}_{t+1}\right\} - \beta\alpha\bar{A}\bar{K}^{\alpha-1}\left(\mathbb{E}_t\left\{\hat{a}_{t+1}\right\} + (\alpha-1)\,\hat{k}_t\right) \tag{15}$$

Log-Linearization of the resource constraint:

$$\underbrace{C_t}_{\bar{C}\hat{c}_t} + \underbrace{K_t}_{\bar{K}\hat{k}_t} = \underbrace{A_t K_{t-1}^{\alpha}}_{\bar{K}^{\alpha}\left(\hat{a}_t+\alpha\hat{k}_{t-1}\right)} + \underbrace{(1-\delta)\,K_{t-1}}_{(1-\delta)\bar{K}\hat{k}_{t-1}}$$

Thus, this equation determines the variations of physical capital as a backward looking process:

$$\hat{k}_t = \bar{K}^{\alpha-1}\left(\hat{a}_t + \alpha\hat{k}_{t-1}\right) + (1-\delta)\,\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t. \tag{16}$$

Log-Linearization of the shock process:
Finally, the log-linearization of the shock process is straightforward:

$$\hat{a}_t = \rho\hat{a}_{t-1} + \eta_t \tag{17}$$

1.7 Calibration

One major advantage of DSGE models is their microfoundations: for instance in our set-up, we have an infinite continuum of households who consume and accumulate capital, our calibration could rely on surveys about their consumption habits. Macroeconomists can use both microeconomic and macroeconomic surveys to calibrate their macroeconomic model and replicate stylized facts. Here our small model aims at replicating the business cycles of an economy through productivity shocks. Table 1 reports the very standard calibration which can be found for the classical growth models.

Parameter	Value	Description
β	0.99	Household discount factor
α	0.40	Share of capital in the production function
δ	0.025	Physical capital depreciation rate
ρ	0.95	AR(1) term productivity shock
σ	0.007	Standard deviation of the productivity shock

Table 1: Calibration of the model

For instance, the depreciation rate of capital is the same as in Solow (1956)'s model, the discount factor replicates a real risk-free rate of 1% annually while the share of capital in the production is comparable to the observed capital share involved in the production process of goods for Western economies (roughly 40% of physical capital and 60% of labour). The following calibration delivers the following steady state using Equation 13 and Equation 14:

Variable	Value
$\bar{K}$	57.7077
$\bar{C}$	3.6213

Table 2: Steady state of capital and consumption

The steady state is (obviously) consistent with the data as both consumption and physical capital are positive, and the stock of physical capital is higher than consumption.

2. Simulations of the model

For an extensive presentation of the Dynare synthax, you can also read the Dynare manual directly. This section is inspired by Callum and Mariano (2013).

2.1 Writing the model in Dynare

We want to write the model into a form Dynare can interpret. In the first part of the Dynare code, we need to define the endogenous variables which appear in the model. There are three equations, so we have three variables: c_t, k_t and a_t. The syntax for writing this in Dynare is var [variable1 variable2, ...]; with the line ended by a semicolumn. For our model, we write:

```
var c k a;
```

Next we write the parameters of the model. In our model, there are four structural parameters (α, β, δ and ρ) plus two steady state variables present in the linear model ($\bar{C}$ and $\bar{K}$), giving six parameters to declare. The syntax for writing this in Dynare is parameters parameter1 parameter2, ... ; with the line ended by a semicolumn. Here we note:

```
parameters  alpha beta delta rho_a Css Kss;
```

Next, we write the exogenous variables in the model. We only have one: η_t. The syntax for writing this in Dynare is varexo variable1 variable2, ... ; with the line ended by a semicolumn. For our set-up, we write:

```
varexo e_a;
```

Finally, we give the parameters values using our calibration in Table 1. The syntax for giving parameters values is parameter = value; which is similar to normal Matlab code. We write:

```
alpha        = 0.40;
beta            = 0.99;
delta            = 0.025;
rho_a        = 0.95;
Kss         = ((1/beta-(1-delta))/alpha)^(1/(alpha-1));
Css         = Kss^alpha - delta*Kss;
```

We now write the equations of the linearised model. Those equations are usually written in the same way as they appear in the paper. The equation section starts with model (linear); and ends with end;. A few syntax considerations:

1. Variables which are multiplied (divided) by a parameter are separated by the parameter by the symbol * (/).

2. Contemporaneous variables do not have a time subscript in the Dynare code. Lagged variables x_{t-i} are written as x(i), while expectational variables $\mathbb{E}_t x_{t+i}$ are written as x(+i).

3. Equations are ended with a semicolumn ;.

The equations in Dynare syntax then are:

```
model(linear);
        [name='Euler equation']
        c = c(+1) - beta*alpha*Kss^(alpha-1)*(a(+1) + (alpha-1)*k);
        [name='Budget constraint']
        k = Kss^(alpha-1)*( a + alpha*k(-1) ) + (1-delta)*k(-1) - Css/Kss*c;
        [name='Productivity shock']
        a = rho_a*a(-1)+e_a;
end;
```

After specifying the equations, we define the shock, and set its standard deviation. We begin the section with a tag shocks; and end it with end;. The syntax for specifying each shock in Dynare is var shock = shock variance;. For the model, the section is:

```
shocks;
        var e_a;  stderr .007;
end
```

Next, we can perform a diagnostic of the model stationarity with respect to the Blanchard-Kanh condition. In Dynare, we add:

```
check;
```

Finally, we solve the model and perform stochastic simulations. To do so, we employ the function stoch_simul(options=value) variable 1 variable 2;. Here we want a 30-period system response of consumption, capital and productivity after a technological shock. We write accordingly:

```
stoch_simul(irf=30) c k a;
```

There many options available for the stoch_simul() function, see directly the Dynare user guide Adjemian et al. (2011) for details about options.

2.2 Simulating the economy

To run the model, the code has to be saved in a .mod file (mine is called log_basicRBC.mod). To run the file, you also need to set Matlab's current directory where your Dynare file is. To initiate Dynare, in Matlab's command line, type:

console

```
dynare log_basicRBC
```

It is possible to insert more argument in the Dynare command (see the manual for details). After pressing enter, you will then see Dynare at work. What it is doing is reading in the .mod contents, recognising the model's details, and solving for the linear rational expectations solution. A key advantage of Dynare compared to other alternatives is that Dynare creates and fills the matrices with the appropriate parameters to solve the linear rational expectation model and prints and saves all results. A lot of informations pops up in Matlab command window, which concerns the stability of the system and the second order moment statistics of the model.

console response

```
Configuring Dynare ...
[mex] Generalized QZ.
[mex] Sylvester equation solution.
[mex] Kronecker products.
[mex] Sparse kronecker products.
[mex] Local state space iteration (second order).
[mex] Bytecode evaluation.
[mex] k-order perturbation solver.
[mex] k-order solution simulation.
[mex] Quasi Monte-Carlo sequence (Sobol).
[mex] Markov Switching SBVAR.

Starting Dynare (version 4.4.3).
Starting preprocessing of the model file ...
Found 3 equation(s).
Evaluating expressions...done
Computing static model derivatives:
- order 1
Computing dynamic model derivatives:
- order 1
- order 2
Processing outputs ...done
Preprocessing completed.
Starting MATLAB/Octave computing.
```

The first bit of printed output Dynare generates is processing information.

Next, Dynare checks whether the Blanchard-Kanh condition is verified.[2]

[2] This is a condition defining the stability condition of the model. As explained in Dynare website, the Blanchard and Kahn conditions state that in order to have a unique stable trajectory you need to have as many eigenvalues larger than 1 in modulus as you have forward looking variables (variables with leads in Dynare; if leads is on two periods as in y(+2), it counts for 2 forward looking variables).

In addition, a matrix that pops up in the computation of the solution must have full rank.

These conditions can be broken two different ways: with eigenvalues more or less numerous than the forward looking

console response

```
EIGENVALUES:
Modulus Real      Imaginary



0.95      0.95     0
0.9691   0.9691 0
1.042    1.042   0
Inf        Inf       0




There are 2 eigenvalue(s) larger than 1 in modulus
for 2 forward-looking variable(s)



The rank condition is verified.
```

Here, as the condition is verified, the simulation can continue.

Next, Dynare generates the variance-covariance matrix of the shocks. The elements on the diagonal correspond to the specified values of the shock's variance. Here, we only have one shock, so the matrix is 1x1:

console response

```
MATRIX OF COVARIANCE OF EXOGENOUS SHOCKS



Variables    e_a
e_a            0.000049
```

Next, we obtain the policy and transition functions. Recall that in a compact form, Dynare computes the solution of the log-linear model (called policy function) as follows:

$$\hat{z}_t = A\hat{z}_{t-1} + B\epsilon_t \tag{18}$$

where $\hat{z}_t = [\hat{c}_t, \hat{k}_t, \hat{a}_t]$ is the vector of the endogenous variables which is a self-determined process based on previous realizations of $\hat{z}_{t-1}$ + the current realization of shocks $\epsilon_t = [\eta_t]$. The matrix A corresponds to the first two rows while B corresponds to the last row of the policy transition matrix.

variables. This leads respectively to nonexistence of a stable trajectory or to an infinity of stable trajectories (indeterminacy). Dynare will only proceed with computation if there exists a unique stable trajectory. You can display the eigenvalues with the "check" command added to the line after "steady" (the command to compute the steady state of a model). If this condition does not hold, Dynare stops processing the code and prints an error message.

console response

```
POLICY AND TRANSITION FUNCTIONS


         c          k          a
k(-1)    0.653600   0.969086   0
a(-1)    0.269310   0.066465   0.950000
e_a      0.283484   0.069963   1.000000
```

We can then rewrite our backward looking process as:

$$\begin{bmatrix} \hat{c}_t \\ \hat{k}_t \\ \hat{a}_t \end{bmatrix} = \begin{bmatrix} 0 & 0.653600 & 0.269310 \\ 0 & 0.969086 & 0.066465 \\ 0 & 0 & 0.950000 \end{bmatrix} \begin{bmatrix} \hat{c}_{t-1} \\ \hat{k}_{t-1} \\ \hat{a}_{t-1} \end{bmatrix} + \begin{bmatrix} 0.283484 \\ 0.069963 \\ 1 \end{bmatrix} [\eta_t]$$

Since consumption $\hat{c}_t$ is only forward looking in the Euler equation, it does not appear in the policy matrix as its row is only a vector of all zeros.

The next section of Dynare output reports the moments of the model, using the model solution and the specified standard errors of the exogenous shocks.

console response

```
THEORETICAL MOMENTS


VARIABLE    MEAN     STD. DEV    VARIANCE
c           0.0000   0.0248      0.0006
k           0.0000   0.0313      0.0010
a           0.0000   0.0224      0.0005
```

Dynare then reports a matrix of theoretical correlations across the endogenous variables. These values are constructed from the variance-covariance matrix of the endogenous variables, which Dynare does not output to the command window.

console response

```
MATRIX OF CORRELATIONS


Variables  c        k        a
c          1.0000   0.9862   0.7516
k          0.9862   1.0000   0.6319
a          0.7516   0.6319   1.0000
```

Next, Dynare shows coefficients of autocorrelation up to the fifth order, that is, the correlation of an endogenous value with p-order lags of itself, $corr(\hat{z}_t, \hat{z}_{t-p})$ where $p = 1, 2, 3, 4, 5$.

console response

```
COEFFICIENTS OF AUTOCORRELATION


Order    1        2        3        4        5
c        0.9965   0.9917   0.9858   0.9788   0.9709
k        0.9992   0.9969   0.9933   0.9884   0.9823
a        0.9500   0.9025   0.8574   0.8145   0.7738
```

Our model is able to reproduce two simple business cycle facts: (i) the persistence of capital is higher than consumption, (ii) consumption is stable over the business cycles.

Finally, Dynare also plots the system impulse response functions (IRF), *i.e.* the response of endogenous variables after the positive realization of each exogenous shock. Since our model only includes one shock, Equation 2.2 draws the response of consumption $\hat{c}_t$, physical capital $\hat{k}_t$ and productivity $\hat{a}_t$ after the realization of 0.7% shock η_t.

Stored in	Description
*oo_*	Structure of the output of stoch_simul function
oo_.exo_simul	Simulated series of the exogenous variables (if simulating).
oo_.endo_simul	Simulated series of the endogenous variables (if simulating).
oo_.dr	Stored information about variables and policy function.
oo_.dr.order_var	The mapping from the declaration order to the decision rule order.
oo_.dr.inv_order_var	The mapping from the decision rule order to the declaration order.
oo_.dr.ghx	The A matrix of the transition function.
oo_.dr.ghu	The B matrix of the transition function.
oo_.steady_state	Values of the endogenous variables' steady-state.
oo_.gamma_y	The variance decomposition outputted to the command window.
oo_.mean	The mean of endogenous variables.
oo_.var	The variance-covariance matrix.
oo_.autocorr	The autocorrelation matrices for endogenous variables; up to 5 lags.
oo_.irfs	Impulse responses of endogenous variables to exogenous shocks.
oo_.irfs.x_y	Impulse response of x to an innovation in y.
oo_.forecast	Forecasts of endogenous variables.
*M_*	Structure with information about the model.
M_.endo_names	The names of endogenous variables in declaration order.
M_.exo_names	The names of exogenous variables in declaration order.
M_.exo_nbr	The number of exogenous variables.
M_.endo_nbr	The number of endogenous variables.
options_	Structure with all the options which Dynare recognises.

Table 3: Matlab's workspace after a stochastic simulation

Exercise 1: playing around with the stoch_simul() function

To do this exercise, first download the Dynare user guide at http://www.Dynare.org/wp-repo/Dynarewp001.pdf and seek for the section related to stoch_simul() function.

1. Plot the response of consumption $\hat{c}_t$ only.
2. Disable the pop-up of the IRF.
3. Disable the console printing.
4. Draw the system response of each endogenous variable using Matlab's plot(·) function (*tips*: to do so, you must find where Dynare has stored the IRF, this information is in the Dynare's User Guide).
5. Generate a sample of 100 artificial series, dropping the first 50 draws:
   (a) Plot each endogenous variable. Does consumption look like the output gap?
   (b) In your opinion, how has Dynare generated these artificial series?
   (c) Compute the theoretical variance of these variables. Do you find the same theoretical variance than the one provided by Dynare? Increase the number of draws and see whether you converge (*tips:* google the function to compute the variance in Matlab).
   (d) Compute the theoretical autocorrelation up to a first order (*tips:* google the function to compute the variance in Matlab).

Exercise 2: extending the simple model

1. Draw the system response of a negative productivity shock. Is the response perfectly symmetric with its previous positive version? Do you find this result realistic?
2. The GDP, Y_t and investment I_t are not (yet) included in the model. Include them and draw their responses to a productivity shock. To do so, log-linearize their equations and find their steady state.
3. Check whether the investment- to-gdp ratio $\bar{I}/\bar{Y}$ and $\bar{C}/\bar{Y}$ are consistent with the data (on average, investment and consumption represent respectively 20% and 60% of the GDP for OECD countries). Why is there such a huge gap? How could you improve the RBC model?
4. Change the calibration to get a non-stationnary system response. What is Dynare telling you in the command window? Does increasing return to scale makes the model non-stationnary?

Exercise 3: extension: habits in consumption

Theoretical macro models typically failed at reproducing the excess return premium on financial markets before the 90s. In asset pricing theory, a new consensus emerged to solve these puzzle through the inclusion of new ingredients in macro-models which would help in solving the puzzle. One of them is the consumption habits, pioneered by Constantinides (1990) and Abel (1990). Typically in real life situations, households have strong consumption habits which materializes in time series by having a very stable consumption over the business cycles, compared to other aggregates such as investment and output. Basically, if the household decides the consume more, it will raise the amount of consumption tomorrow. However, there is few evidence about such behaviour at a micro-level, suggesting that consumption habits is probably a shortcut for a more complex explanation of consumption inertia.

1. Let us assume first *external* habits. These external habits simply change how households are enjoying consumption in their utility function:

$$\mathcal{U}\left(C_{jt} - hC_{t-1}\right) \tag{19}$$

   where $h \in [0, 1)$ is the degree of consumption habits. Here, it is critical to notice that these habits are depending on the aggregate amount of consumption C_{t-1} (i.e. there is no subscript j). So basically, the amount of aggregate consumption is not a control variable when households maximizes utility. This is refered to as *catching up with the jonese* as people will copy the behaviour of the majority of consumers.[a]

   Here, the steady state is not affected by this new addon, only the FOC does:

$$\frac{\mathcal{U}'\left(C_{jt} - hC_{t-1}\right)}{\mathbb{E}_t\left\{\mathcal{U}'\left(C_{jt+1} - hC_t\right)\right\}} = \beta\mathbb{E}_t\left\{r_{t+1}\right\} \tag{20}$$

   Find the vale of $\mathcal{U}'\left(C_{jt} - hC_{t-1}\right)$, log-linearize it and include it in your model. How persistent is now consumption?

2. Let

[a] See Abel (1990) the story about the Joneses.

Exercise 4: demand shocks

Assume now that our economy has a government. The government finances public spending by charging a lump-sum tax T_{jt} on households. The total amount of public spending reads as, $P_t G_t$. The balance sheet of government writes:

$$P_t G_t = \textstyle\int_0^1 T_{jt} \mathrm{d}j \quad (=T_t). \tag{21}$$

Since the economy is populated by an uniform household, we can write T_t directly. The budget constraint of the representative household is now:

$$C_{jt} + I_{jt} = A_t K_{jt-1}^{\alpha} - \frac{T_{jt}}{P_t}, \tag{22}$$

while the resource constraint/the GDP through the expenditure approach is now given by:

$$Y_t = C_t + I_t + G_t. \tag{23}$$

The decision of increasing/decreasing spendings is decided exogenously by the governement. In absence of an explicit modelling of political cycles, the spending is assumed to vary around a fixed value of output:

$$G_t = \bar{G}\varepsilon_t^G = g^y \bar{Y} \varepsilon_t^G, \tag{24}$$

where spending follows an $AR(1)$ process:

$$\log\left(\varepsilon_t^G\right) = \rho_G \log\left(\varepsilon_{t-1}^G\right) + \eta_t^G, \tag{25}$$

with $0 \leq \rho_G < 1$ is the autocorrelation coefficient of the demand shock and η_t^G is an innovation distributed normally with zero mean and finite variance σ_G^2 such that $\eta_t^G \sim \mathcal{N}(0, \sigma_G^2)$. Parameter g^y is the spending-to-GDP ratio. We borrow the values of Smets and Wouters (2007) and assume that $\rho_G = 0.97$, $\sigma_G = 0.053$ and $g^y = 0.20$. Implicitely in steady state, the shock is normalized to one $\bar{\varepsilon}^G = 1$.

1. Compute the new steady state of the model. *Tips:* (i) define a new parameter, $g^y = 0.2$, which affects the new steady state; (ii) since the budget is balanced, you can substitute real taxes T_t/P_t by spending G_t.

2. Let's now calibrate the demand shock process, show the IRF after the realization of the demand shock.

3. According to Keynesian economics, is this shock consistent with this theory? Why do households are "Ricardian"? Is this behaviour observed in the data?

4. Analyse the variance decomposition, what is the main driver of output?

5. Is this RBC model now able to replicate another business cycle fact which says output is more volatile than consumption?

We consider a time-preference shock which is also considered as a demand shock, *i.e.* the household discount factor β is affected by an exogenous component ε_t^C. Basically, the representative household temporary offsets her consumption smoothing by increasing (decreasing) its spending for a positive (negative) realization of this time-preference shock, as she becomes transtory more patient (impatient). The new objective function of the household reads as:

$$\max_{\{C_{jt},K_{jt}\}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \varepsilon_{t+\tau}^C \log\left(C_{jt+\tau}\right) \tag{26}$$

where the time preference shock reads as:

$$\log\left(\varepsilon_t^C\right) = \rho_C \log\left(\varepsilon_{t-1}^C\right) + \eta_t^C. \tag{27}$$

We assume the following calibration as in Smets and Wouters (2007) for this shock: $\bar{\varepsilon}^C = 1$, $\rho_C = 0.22$ and $\sigma_C = 0.023$.

6. How is the Euler equation affected by this shock? Take it in logs.
7. Simulate the system response, what is the effect of the shock on the economy? Analyse the variance decomposition.
8 Using simultaneously all the shock, analyse the forecast error variance decomposition of the model up to 1, 2, 4, 10 and 100 quarters (Find the function in Dynare's userguide). Which shocks is driving output in the short and in the long run?

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Appendices

A. Installing Dynare

Dynare is a all-in-one matlab package which allows you to simulate both stochastic and deterministic models, make estimations, approximate the policy function up to higher order approximations, etc. This package is easy to install:

1. Go to the Dynare website: http://www.Dynare.org/download/Dynare-stable and install the Dynare package.

2. Once Dynare is installed, you must indicate to matlab where to load Dynare files each time Matlab starts. Find the "set path" option in Matlab.

3. Click on "Add Folder"

4. Go to your Dynare folder, and select the /matlab/ folder inside, and click on "Select that Folder". For instance on my PC, the latest Dynare version is installed in C:\Dynare\4.4.3\.

5. Save your modification, click on the "Save" button.

Now, Dynare should be called when you type "Dynare" in the Matlab console.

B. A note on log-linearization

For any function $f(x_t)$ with corresponding steady state $f(\bar{x})$, a Taylor expansion up to first order is given by:

$$f(x_t) \simeq f(\bar{x}) + f'(\bar{x})\,(x_t - \bar{x}) \tag{28}$$

Expressed in terms of deviation from the steady state, $f(x_t) - f(\bar{x})$, the expansion can be rewritten in the following way:

$$\begin{aligned} f(x_t) - f(\bar{x}) &\simeq f'(\bar{x})\,(x_t - \bar{x}) \\ &\simeq f'(\bar{x})\bar{x}\frac{(x_t - \bar{x})}{\bar{x}} \\ &\simeq f'(\bar{x})\bar{x}\log(\frac{x_t}{\bar{x}}) \\ &\simeq f'(\bar{x})\bar{x}\hat{x}_t \end{aligned}$$

where $\hat{x}_t = \log(x_t/\bar{x})$ is the percentage variations of x_t around its steady state.

For a two variable fonction, the approximation reads as:

$$f(x_t, y_t) - f(\bar{x}, \bar{y}) \simeq f'_x(\bar{x}, \bar{y})\bar{x}\hat{x}_t + f'_y(\bar{x}, \bar{y})\bar{y}\hat{y}_t \tag{29}$$

We can them summarize these general approximations by some tricks that we can use to loglinearise quickly a model.

Theorem 1 *Log-linearisation of a variable*

$$x_t \longrightarrow \bar{x}\hat{x}_t$$

Theorem 2 *Log-linearisation of a product of two variables*

$$x_t y_t \longrightarrow \bar{x}\bar{y}(\hat{x}_t + \hat{y}_t)$$

Theorem 3 *Log-linearisation of a ratio of two variables*

$$\frac{x_t}{y_t} \longrightarrow \frac{\bar{x}}{\bar{y}}(\hat{x}_t - \hat{y}_t)$$

Theorem 4 *Log-linearisation a variable with exponent*

$$x_t^a \longrightarrow \bar{x}^a a \hat{x}_t$$

C. Solution of a linear rational expectation model

Une fois le modèle sous forme log-linéarisée avec son état stationnaire connu, on définit le vecteur y_t des variables endogènes et le vecteur z_t pour les variables exogènes (les chocs) au modèle. Le modèle peut-être réécrit avec F, G, H, L et M des matrices,

$$0 = E_t \left[\; Fy_{t+1} + \quad Gy_t + \quad Hy_{t-1} + \quad Lz_{t+1} + \quad Mz_t \; \right]$$

Pour trouver la solution du problème, il faut déterminer les matrices P et Q, de la loi de mouvement suivante,

$$y_t = Py_{t-1} + Qz_t$$

dont l'équilibre décrit par ce processus est stable. Ce problème peut parfois être difficile à résoudre car les racines de P peuvent être complexes et gourmandes en calculs, en particulier si le modèle est large. La solution de la version linéaire de l'économie est habituellement plus facile à trouver si l'on coupe en deux avec d'un côté un vecteur de variables d'état, v_t, et de l'autre un vecteur de variables de saut, w_t[3]. Ce nom de variables de saut fait référence aux dynamiques de point-selle telles que le système est stable et convergent le long de l'extremum de la selle. Habituellement, les variables de contrôle suivent naturellement l'extremum, mais les valeurs des autres variables ont besoin d'être (instantanément) sautées afin d'obtenir un système stable sur l'extremum de la selle Uhlig et al. (1995).

En séparant les équations qui incorporent des anticipations (y_{t+1} ,w_{t+1}) de celles qui n'en ont pas, la version linéaire du modèle peut être écrite comme,

Sans anticipation	$0 = Ay_t + By_{t-1} + Cw_t + Dz_t$
Avec anticipations	$0 = E_t \left[Fy_{t+1} + Gy_t + Hy_{t-1} + Jw_{t+1} + Kw_t + Lz_{t+1} + Mz_t \right]$
Loi de mouvement	$z_{t+1} = Nz_t + \varepsilon_{t+1}$, tel que $E\left[\varepsilon_{t+1}\right] = 0$

La solution est un ensemble de matrices P, Q, R et S, qui décrivent la loi de mouvement de l'équilibre de la forme,

$$y_t = Py_{t-1} + Qz_t \text{ et } w_t = Rw_{t-1} + Sz_t$$

Ces matrices sont calculés en ne gardant que les valeurs propres stables **?**. On note que pour notre modèle, la matrice C est de rang plein et inversible, C^{-1}. A partir du corollaire de Uhlig, si la loi de mouvement de l'équilibre P existe, il doit remplir,

$$\begin{aligned} &0 = \left(F - JC^{-1}A\right) P^2 - \left(JC^{-1}B - G + KC^{-1}A\right) P - KC^{-1}B + H \\ &R = -C^{-1}\left(AP + B\right) \end{aligned}$$

[3] Appelé aussi vecteur de contrôle.

Q doit satisfaire l'équation,

$$vec\left(Q\right)=\frac{vec\left(\left(JC^{-1}D-L\right)N+KC^{-1}D-M\right)}{\left(N\otimes\left(F-JC^{-1}A\right)+I_k\otimes\left(JR+FP+G-KC^{-1}A\right)\right)}$$

et

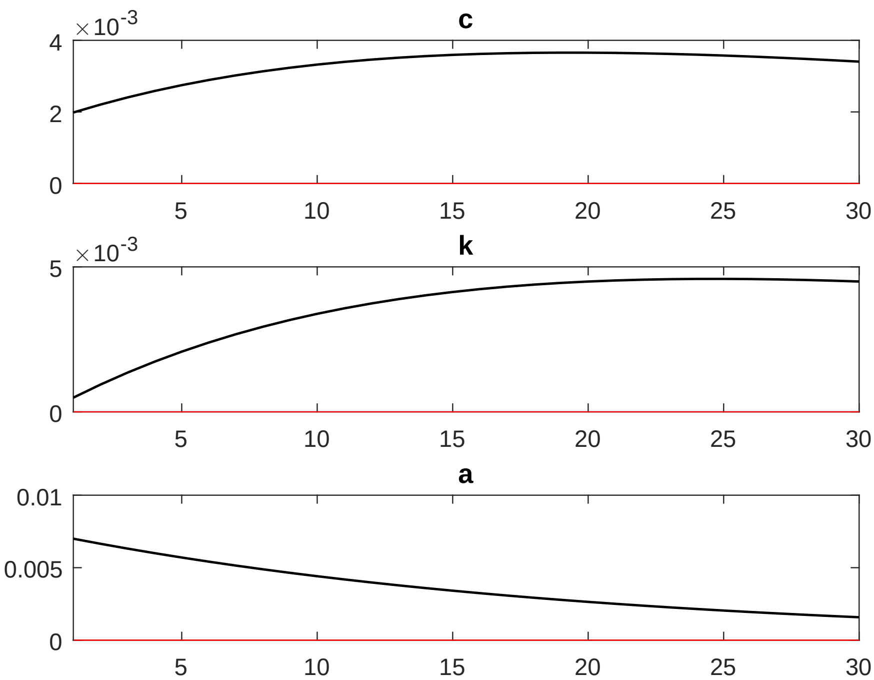
$$S=-C^{-1}\left(AQ+D\right)$$

La matrice identité I_k est de dimension $k\times k$, où k est le nombre de colonnes dans la matrice Q. Par ailleurs, $\otimes$ est le produit de Kronenker[4]. L'ensemble de ces calculs sont automatisés sous Dynare.

[4] Le produit de Konenker est simplement :

$A=\begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{bmatrix}$ et $B=\begin{bmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{bmatrix}$

$$\begin{aligned} A\otimes B &= \begin{bmatrix} A_{1,1}*B & A_{1,2}*B \\ A_{2,1}*B & A_{2,2}*B \end{bmatrix} \\ &= \begin{bmatrix} A_{1,1}*B_{1,1} & A_{1,1}*B_{1,2} & A_{1,2}*B_{1,1} & A_{1,2}*B_{1,2} \\ A_{1,1}*B_{2,1} & A_{1,1}*B_{2,2} & A_{1,2}*B_{2,1} & A_{1,2}*B_{2,2} \\ A_{2,1}*B_{1,1} & A_{2,1}*B_{1,2} & A_{2,2}*B_{1,1} & A_{2,2}*B_{1,2} \\ A_{2,1}*B_{2,1} & A_{2,1}*B_{2,2} & A_{2,2}*B_{2,1} & A_{2,2}*B_{2,2} \end{bmatrix} \end{aligned}$$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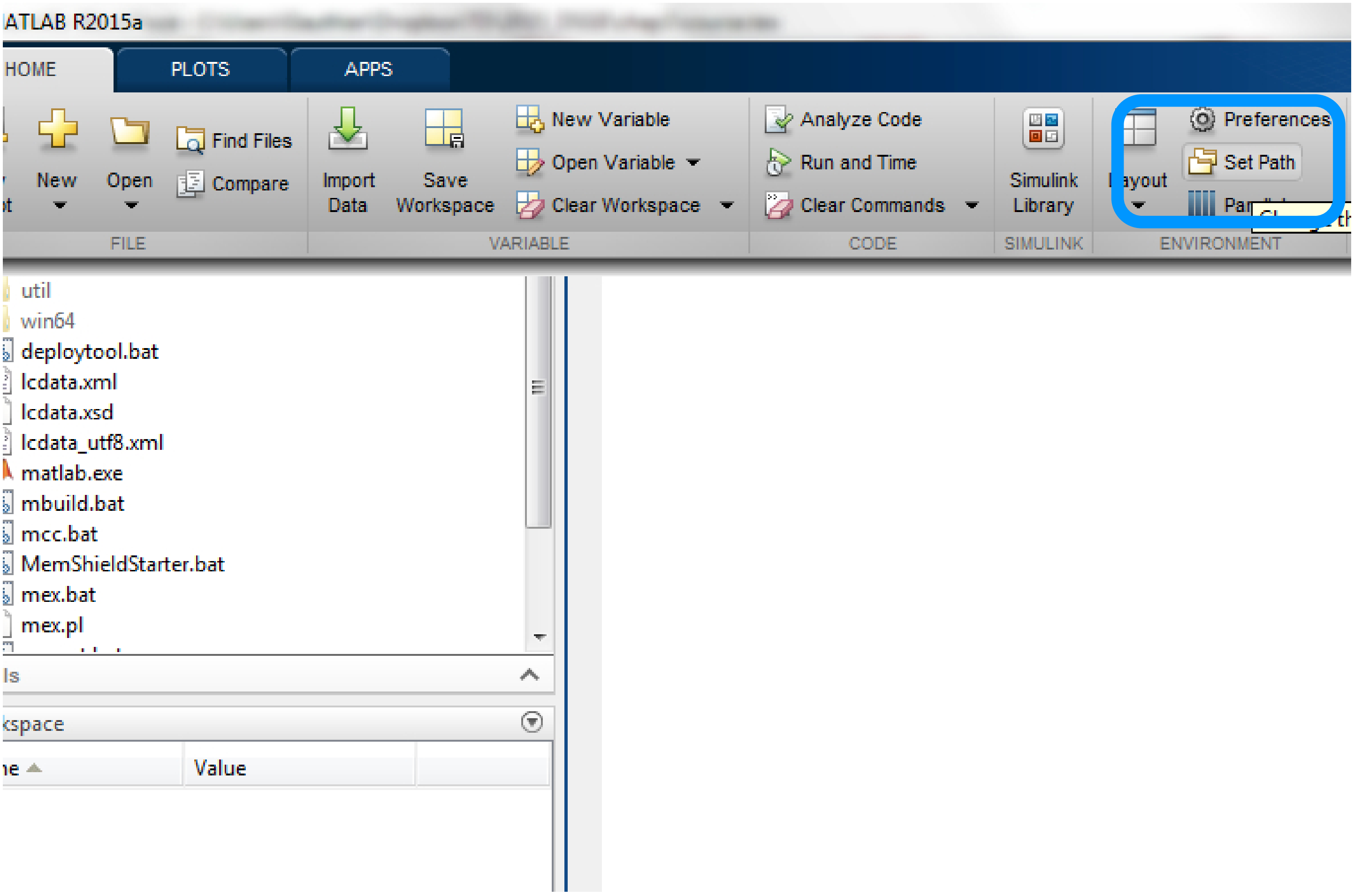
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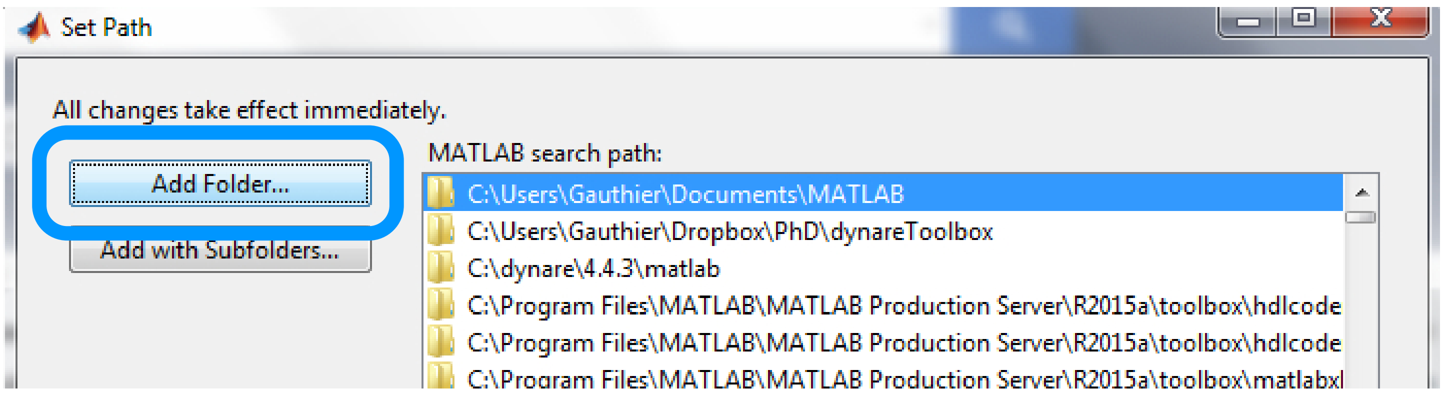
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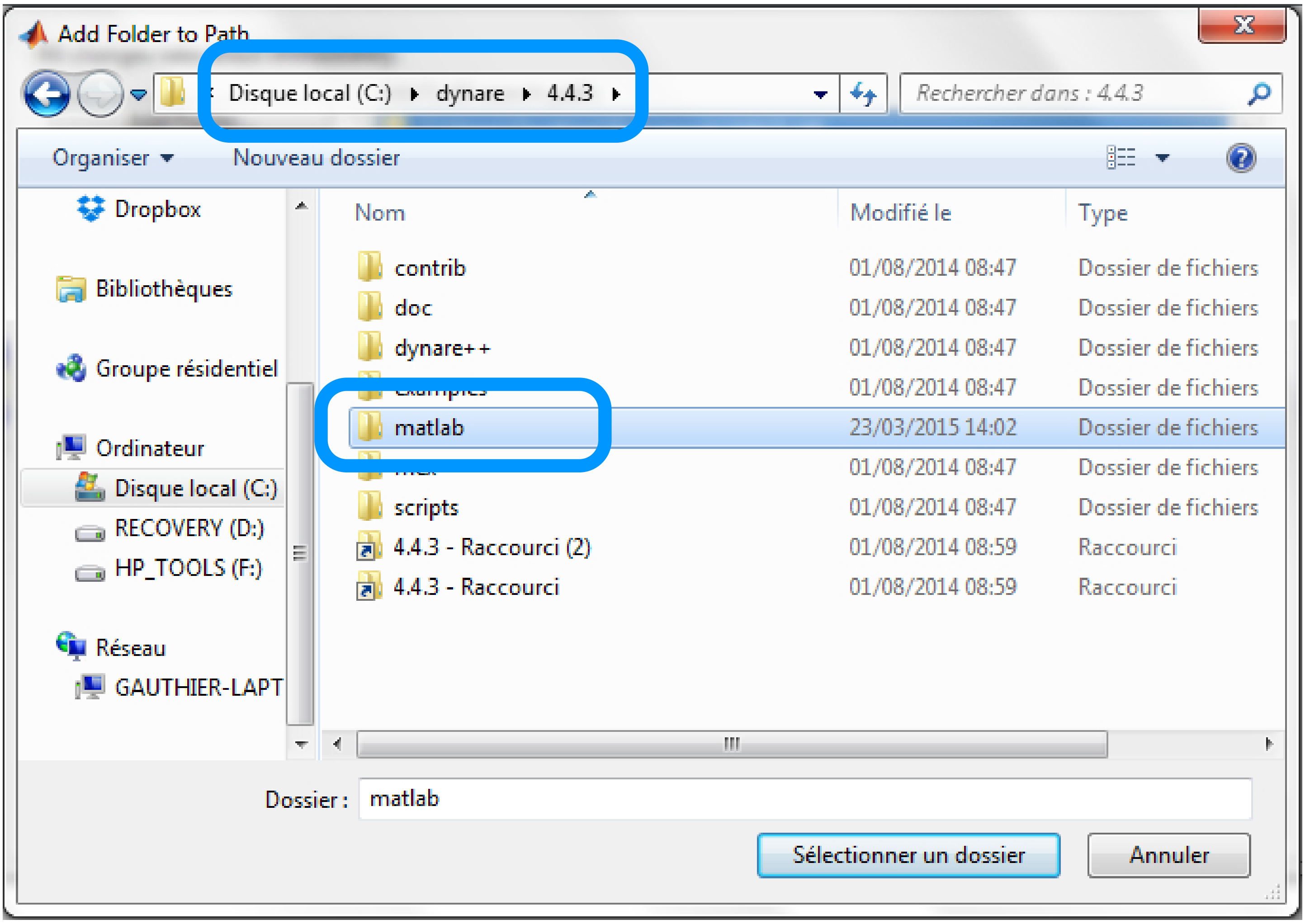
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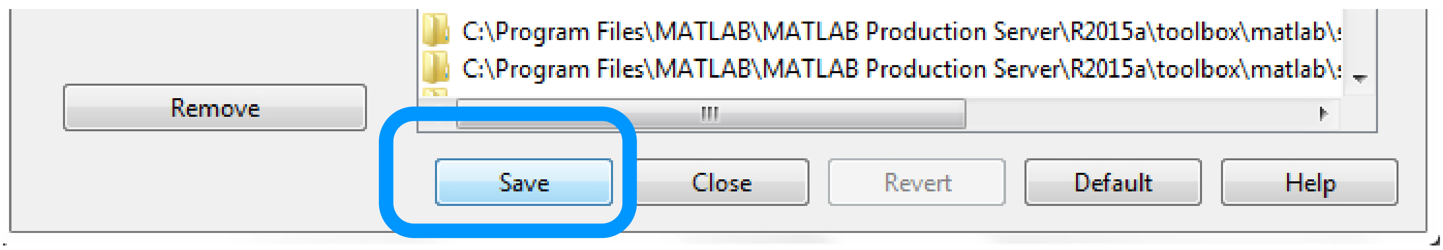
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